

Library Management System – Project Guidelines

1. Project Objective

The objective of this project is to build a **Library Management System** that manages books and members.

The system allows users to **view books, register members, borrow books, and return books**, while storing records in files.

2. Functional Requirements

Book Management

- Add new books

- View all books

- Update book availability

- Store book records in a file

Member Management

- Register members

- View member details

- Track borrowed books

Borrow & Return

- Borrow a book

- Return a book

- Prevent borrowing if the book is unavailable

3. System Models

Book Model

Represents a book in the library.

Attributes:

- Book ID

Title

Author

Category

Availability Status

Responsibilities:

Store book details

Track availability

Provide book information

Example:

Book ID: 101

Title: Python Programming

Author: John Smith

Status: Available

Member Model

Represents a library member.

Attributes:

Member ID

Name

Email

Borrowed Books List

Responsibilities:

Store member details

Track borrowed books

Example:

Member ID: M001

Name: Ravi Kumar

Borrowed Books: 101

4. System Workflow

Start application (Menu-driven system)

Menu Options:

- 1 View Books
- 2 Add Book
- 3 Register Member
- 4 Borrow Book
- 5 Return Book
- 6 Exit

View Books → display all books from file

Register Member → store member details in file

Borrow Book → verify book & member → update status

Return Book → update book availability

library_services.py – Documentation (Without Code)

Overview

The **library_services module** contains the **core business logic** of the Library Management System. It is responsible for handling operations such as **managing books, registering members, borrowing books, returning books, and retrieving library records**.

This module acts as the **service layer** between the **main application (user interface/menu system)** and the **data storage layer (files containing book and member data)**.

Responsibilities of the Module

The **library_services** module performs the following tasks:

Manage library books

Register and maintain member records

Handle book borrowing and returning

Track book availability

Update book and member data in files

Validate input data

Handle exceptions during operations

5. File Handling

Two files should be maintained.

Books File

Stores book records.

Fields:

Book ID

Title

Author

Category

Availability

Members File

Stores member details.

Fields:

Member ID

Name

Email

Borrowed Books

6. Exception Handling

The system should handle errors such as:

Book Not Found – when an invalid book ID is entered

Book Already Borrowed – when a borrowed book is requested again

Member Not Found – when an unregistered member borrows a book

File Not Found – when data files are missing

7. Project Structure

```

library_management_system
├── main.py
├── models
│   ├── book.py
│   └── member.py
├── services
│   └── library_service.py
├── utils
│   └── file_handler.py
├── exceptions
│   └── custom_exceptions.py
├── data
│   ├── books.csv
│   └── members.csv
├── tests
│   └── test_library.py
└── README.md

```

Library Management System Test Suite

Test Case	Test Scenario	Precondition	Input Data	Expected Output
TC01	View all books	Books data file contains	Select View Books	System displays all books (ID, Title, Author, Category, Availability).
TC02	Add a new book	System running; file	Book ID, Title, Author,	Book added to file; message " Book added successfully " displayed.
TC03	Register a new member	System running	Member ID, Name, Email	Member saved to file; message " Member registered successfully "
TC04	Borrow available	Member exists; book is	Member ID, Book ID	Status changes to Borrowed ; message " Book borrowed "
TC05	Borrow already	Book status is Borrowed	Member ID, Book ID	System displays " Error: Book is already borrowed ".
TC06	Borrow book (Invalid ID)	Member exists; Book ID is	Member ID, Invalid Book	System displays " Error: Book not found ".
TC07	Borrow book (Invalid	Book exists; Member ID is	Invalid Member ID,	System displays " Error: Member not found ".

TC07	Borrow book (Invalid	Book exists; Member ID is	Invalid Member ID,	System displays "Error: Member not found" .
TC08	Return borrowed	Member currently has	Member ID, Book ID	Status changes to Available ; message "Book returned"
TC09	Return book not borrowed	Member didn't borrow that	Member ID, Book ID	System displays "Error: Book not borrowed by this member" .
TC10	Data file missing	Books/ Members file	View Books or Borrow	System displays "Error: Data file not found" .

Sample Input and Output

Library Management System

Initial Data (Stored in File)

Books File

Book ID	Title	Author	Category	Status
101	Python Programming	John Smith	Programming	Available
102	Data Science Handbook	Jane Miller	Data Science	Available
103	Machine Learning	Andrew Ng	AI	Available

Program Execution

Menu Display

Output

```
===== Library Management System =====
```

1. View Books
2. Register Member
3. Borrow Book
4. Return Book
5. Exit

Enter your choice:

Scenario 1: View Books

Input

Enter your choice: 1

Output

Available Books

Book ID : 101

Title : Python Programming

Author : John Smith

Status : Available

Book ID : 102

Title : Data Science Handbook

Author : Jane Miller

Status : Available

Book ID : 103

Title : Machine Learning

Author : Andrew Ng

Status : Available

Scenario 2: Register Member

Input

Enter your choice: 2

Enter Member ID: M001

Enter Member Name: Ravi

Enter Email: ravi@gmail.com

Output

Member registered successfully.

Scenario 3: Borrow Book

Input

Enter your choice: 3

Enter Member ID: M001

Enter Book ID: 101

Output

Book borrowed successfully.

Scenario 4: Borrow Same Book Again

Input

Enter your choice: 3
Enter Member ID: M002
Enter Book ID: 101

Output

Error: Book is currently not available.

Scenario 5: Return Book

Input

Enter your choice: 4
Enter Member ID: M001
Enter Book ID: 101

Output

Book returned successfully.

Scenario 6: Borrow Non-Existing Book

Input

Enter your choice: 3
Enter Member ID: M001
Enter Book ID: 999

Output

Error: Book not found in the library.

Scenario 7: Exit Program

Input

Enter your choice: 5

Output

Thank you for using the Library Management System.

Sample Updated Data After Borrowing

Books File

Book ID	Title	Author	Category	Status
101	Python Programming	John Smith	Programming	Borrowed
102	Data Science Handbook	Jane Miller	Data Science	Available
103	Machine Learning	Andrew Ng	AI	Available